The following are screenshots from past exemplary student group project submissions. The group project specification may differ this year from that of previous years. Note that the screenshots may not cover all topics required for this year's project.

BioNTech – Market Overview

Porter's Five Forces & Risks – Biotechnology Industry

Competitive Rivalry (High)
- **Mature industry:** BioNTech built pioneer leader image
- **High barriers to exit** (long development cycles, high sunk costs)

⚠ **Growing biotech sector:** AI, mRNA, and personalised medicine fuel competition

Supplier Power (Moderate)
- **Reliance on specialised suppliers** (mRNA raw materials, AI-driven platforms)
- Limited number of high-quality suppliers meaning **bargaining power**
- **Integration risk** if in-house manufacturing

Customer Power (Moderate)
- **Governments, healthcare providers, and insurers** are key buyers
- **Switching costs are high** for innovative treatments (personalised cancer therapies)
- **Price sensitivity varies:** High for vaccines (government bulk purchases), lower for breakthrough oncology treatments

Threat of Substitutes (Low/Moderate)
- **Few direct substitutes** for cutting-edge therapies (mRNA, cell therapy)
- Slow shift from traditional treatments (chemotherapy) to **personalised medicine**
- **Biosimilars & generics** once patents expire

Threat of New Entrants (Low)

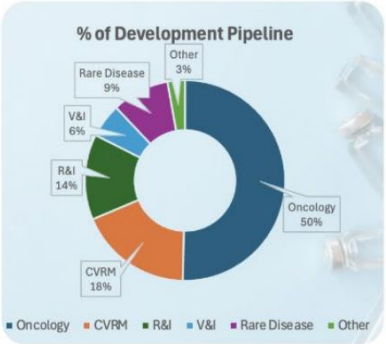
⚠ High **R&D costs, stringent regulations (FDA)**, and **complex manufacturing balancing financial sustainability and innovation**
- **Strong IP protection** for biotech innovations (patents, exclusivity agreements)
- **Established distribution channels & partnerships** needed to scale

⚠ **Regulatory Uncertainty**: Long approval & clinical trial risks with 90% failure

⚠ **Pipeline Dependence**
Crucial oncology approvals to sustain revenue post-COVID. BNT fully relies on Comirnaty, depends on partnerships for market access, making it more cyclical compared to Big Pharma

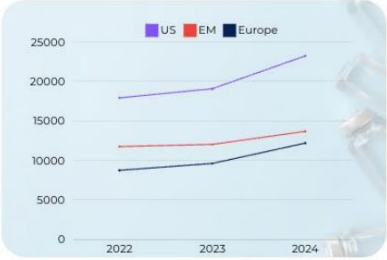
AstraZeneca

Core Competencies



- Increased **Alliance and Collaboration Revenue** by 55% and 56% in 2024

Eg: Oxford Covid Vaccine, Fusion Oncology Biopharma



Key Risks & Mitigation Strategies

AstraZeneca

Risks		Mitigation
1. Product Pipeline and R&D Risk	Failure or delay in drug development and approvals, regulatory risks	Prioritizing high-potential R&D projects, strengthening pipeline through acquisitions, licensing, collaboration
2. Supply Chain and Manufacture Risks	Failure to maintain supply of compliant, high-quality medicines, dependence on third-party suppliers and contract manufacturers	Establishing new manufacturing facilities , implementing dual sourcing of essential raw materials , business continuity planning
3.Commercialization and Market Risks	Pricing and reimbursement pressures, increased competition, market access and affordability	Dynamic pricing models and innovative reimbursement agreements, expanding market reach, investing in personalized medicine and high-margin specialty drugs
4. Legal, Compliance, and IP Risks	Litigation and government investigations, Intellectual property (IP) risks	Strong patent defence strategy, regular audits, engagement with regulatory authorities
5. Financial & Macroeconomic Risks	Geopolitical and macroeconomic volatility,, Access to capital and financial risks, FX risk	Hedging currency risks and fluctuations, maintaining a diversified revenue stream , strategic capital allocation and cost management initiatives
6.Cybersecurity and Data Privacy Risks	Cyber threats and IT disruptions, failure to manage data and AI in compliance with regulations	Implementing an enterprise-wide cybersecurity framework, employee cybersecurity awareness programs, strict data governance policies
7.Environmental and Sustainability Risks	Climate-related disruptions, failure to meet sustainability targets	Reduction plans for Scope 1, 2, and 3 net-zero emissions, investment in renewable energy and sustainable production methods, climate risk assessments integrated into business continuity planning

Identified Risks So Far – Expanding to Industry Challenges in Tandem

Risk classification: High Medium Low

Strategic Risk[1,2]

Failure To Identify Business Opportunities
- Ability to identify targets and partnerships.
- Third party dependencies in value chain.

Opella Separation Risk
- Conditions for completion may not be met.
- Unintended negative business impact.

Operational Transformation Risk
- Employee resistance for change.

Globalization Challenges
- Emerging market risks: economic volatility, currency fluctuations, and regulatory hurdles.
- Long-term epidemics may disrupt operations.
- Regulatory differences (EU vs. US) limit collaborations.
- China (6.5% of 2024 net sales) is a major but risky market.

Digitalization & AI
- Dependency on data access, quality & evolvement of AI.
- Competition from tech firms with advanced capabilities.

Industry Risk

Threat of substitutes

- **Revenue and market share erosion from generics** already affecting flagship products (e.g., Lantus, Plavix, Aubagio)
- Availability of **alternative treatments** (e.g., novel therapeutics, emerging biotechnologies, non-pharma interventions)
- **Economic pressures** (e.g., inflation, cost-cutting in healthcare systems) pushing demand toward cheaper substitutes
- Government policies favouring **low-cost** generics or alternatives

BP[3] of suppliers

- Sanofi's in-house production of active ingredients and main drug products **reduces supplier dependency** and enhances cost control.
- Reliance on third-party suppliers for **specific ingredients** adds supply risk

Degree of Rivalry

- **Fierce R&D race** for new drugs and **strong competition** even among patented medicines.
- **Generics & biosimilars erode market** share post-patent expiry.
- **Parallel trade** in EU countries. Additional price pressures.

BP[3] of buyers

- **Limited buyer influence** when product is unique or protected by IP
- **Availability of alternatives** (e.g., generics or competing products)
- **Large-scale purchasers** exert pricing pressure

Threat of new entrants

- **High barriers to entry** due to strict regulation, IP protection, and high R&D costs
- The introduction of **generic drugs and biosimilar** manufacturers leads to a rapid and significant decline in revenues. Similarly for emerging tech-driven pharma.
- **New biotech firms** (e.g., mRNA) challenge traditional players.
- **Capital intensity** and **long development timelines** deter small entrants.

sanofi

1. ESG Risks as part of the strategy to be examined separately | 2. Only highest strategic risk considered | 3. Bargaining Power
Sources: Own Analysis, Company Website, Annual Report

Non-Adjusted Accounting Policies (1/2)

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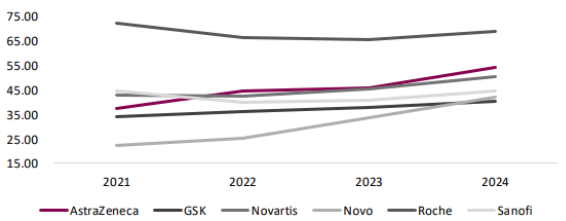
Consistent Revenue Growth Trend ($bn)



Depreciation Policy - Useful Life (yrs) and Method

Company	Buildings	Plant & Equipment	Depreciation Method
AstraZeneca	**10-50**	**3-15**	**Straight Line**
GSK	20-50	3-20	Straight Line
Novartis	20-40	3-20	Straight Line
Novo	10-50	3-25	Straight Line
Roche	10-50	3-15	Straight Line
Sanofi	15-40	3-20	Straight Line
Average Max	**46.7**	**19.2**	
Average Min	**14.2**	**3.0**	

Immaterial Receivable Turnover Ratio Difference

Company	2021	2022	2023	2024
AstraZeneca	4.53	4.43	4.08	4.35
GSK	4.24	4.83	5.35	5.54
Novartis	5.27	5.25	5.99	6.93
Novo	4.16	3.72	4.00	4.32
Roche	5.91	5.17	4.86	5.12
Sanofi	5.03	4.50	4.46	5.15
Average	4.86	4.65	4.79	5.23
AstraZeneca Δ	**-0.33**	**-0.22**	**-0.71**	**-0.88**

Depreciation Policy Analysis

- AstraZeneca's **useful life** assumptions are within the **industry range**, with no major deviations
- The **straight-line method** is consistent across **all peers**, ensuring comparability
- Despite Plant & Equipment being in the lower end of peer range, differences in equipment types, usage intensity, or maintenance policies may **justify a shorter useful life**
- AstraZeneca does **not** report an **estimated salvage value**, which is common in the industry
- **No** clear **evidence** that AstraZeneca's policy **materially distorts financials**
- The impact on **depreciation expense** and overall valuation would likely be **minimal**

AstraZeneca

Sources: 2024-2021 Annual Reports of AstraZeneca, GSK, Novartis, Novo Nordisk and Sanofi

Non-Adjusted Accounting Policies (2/2)

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R&D Capitalisation

- **Capitalisation permitted under IFRS**, but pharma companies rarely apply it due to high uncertainty before regulatory approval; most peers, including AstraZeneca, expense nearly all R&D
- **Capitalised R&D is immaterial across the sector –** with the exception of GSK which is an outlier. While this may reflect differences in development stage or accounting policy, adjustments are more relevant when analysing GSK specifically, not for normalising peer comparisons
- Forward-looking normalisation of R&D investment belongs in forecasting models, not accounting restatements – especially given AstraZeneca's **transparent and conservative treatment** with no evidence of earnings management

Operating Leases Capitalisation

- **Capitalisation of operating leases is permitted under IFRS 16**, but most pharmaceutical companies, including AstraZeneca, apply **the recognition exemptions** for short-term leases (less than 12 months) and low-value leases (under $5,000)
- Short-term and low-value leases **are expensed consistently across the sector**, ensuring comparability and since these leases are not material, capitalising them would not meaningfully change AstraZeneca's financial position
- Normalisation of lease accounting should focus on **material lease obligations**. These leases are typically for small, temporary assets that don't affect long-term financial performance. Keeping them expensed **simplifies financial reporting** and **better reflects their limited economic impact**

Capitalised R&D as % of expensed R&D

Company	2023	2024
AstraZeneca	**0.00%**	**0.00%**
GSK	5.67%	3.95%
Novartis	1.00%	0.88%
Novo	0.00%	0.00%
Roche	Not disclosed	Not disclosed
Sanofi	Not disclosed	Not disclosed

Other Non-Adjusted Policies

The Group **applies IAS 36 "Impairment of Assets"** to assess the **carrying values of non-financial assets**, excluding inventories and deferred tax assets and it results in line with industry policies

Impairments for Intangible Assets and PP&E are strictly **correlated** to the outcome of **clinical trials**, so they are naturally **volatile** and reflect **operational risk** rather than accounting discretion

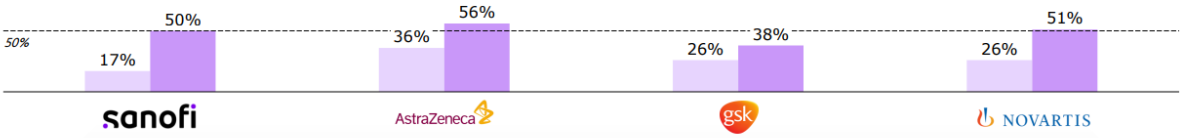
Goodwill was **not impaired**, consistent with **annual testing requirements under IFRS** and in absence of observable triggers

Sources: 2024-2021 Annual Reports of AstraZeneca, GSK, Novartis, Novo Nordisk and Sanofi

Considered Adjustments – Significant Intangible Assets for all companies considered

Intangible Assets (excl. Goodwill) | Intangible Assets (incl. Goodwill)

%-Intangible assets of total assets



Intangible assets were a **key focus for potential adjustments** due to their **material impact on financial statements**, as they represent a **significant portion of the companies' total assets**, are subject to impairment risks, and can vary based on accounting policies.

Resulting adjustment focus

Detailed comparison and treatment overview in Valuation Model

Considered Areas	Adjustment?	Justification	Adjustment
1 Amortisation rate of intangible assets other than goodwill	☐	Sanofi's amortisation rate has been stable over time and is in line with peers	-
2 Impairment of intangible assets other than goodwill	☑	Material impact on income statements and substantial variance across time	Reclassification
3 Rate of additions to intangible assets	☐	Different development schedules leading to different capitalization timings	-
4 Accruals and Receivables Sanofi	☐	Significant changes explainable by business requirements at the time	-

sanofi Sources: Annual Reports selected companies

Reclassification of Impairment Charges to correctly reflect as non-recurring expenses

Detailed calculation in Valuation Model

Basis for Analysis

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Impairment included in

Spot Exchange Rate €/€	1	1	1	1	1	1	1	1
Impairment of Intangible Assets	-295,00	-718,00	-3.604,00	-330,00	-192,00	429,00	-896,00	-248,00

separate line item

Classified as non-recurring expense[1]

AstraZeneca

Impairment included in

Spot exchange rate ($)	0,83	0,87	0,89	0,82	0,88	0,94	0,90	0,97
Net impairment charges - reclassifaction to non-recurring								
Research and development expenses	101,00	539,00	609,00	55,00	1,464,00	95,00	417,00	1,065,00
Selling, general and administrative expense	390,00	144,00	427,00	185,00	621,00	129,00	17,00	509,00
Other operating income and expense			-3,00					

R&D | SG&A | Other income/ expenses

Reclassification from OPEX to non-recurring expenses[1]

gsk

Impairment included in

Spot exchange rate (£)	1,13	1,11	1,18	1,12	1,19	1,13	1,15	1,21
Net impairment charges - reclassifaction to non-recurring								
Cost of sales	400,00	69,00	34,00	21,00		2,00		
Selling, general and administration	2,00	19,00	43,00	17,00	65,00	66,00	18,00	6,00
Research and development	278,00	46,00	49,00	219,00	373,00	299,00	399,00	311,00

COGS | R&D | Other income/ expenses

Reclassification from OPEX to non-recurring expenses[1]

NOVARTIS

Impairment included in

Spot exchange rate ($)	0,83	0,87	0,89	0,82	0,88	0,94	0,90	0,97
Net impairment charges								
Cost of goods sold	92,00	488,00	85,00	377,00	360,00	314,00	310,00	-9,00
Research and development	594,00	167,00	1.078,00	523,00		953,00	2.737,00	500,00
Other income							-10,00	
Other expense						398,00	11,00	

COGS | R&D | Other income/ expenses

Reclassification from OPEX to non-recurring expenses[1]

sanofi 1. Only affect the classification in the Income Statements
Sources: Own calculations based on annual reports published by mentioned companies

We defined new industry-specific ratios to assess the competitiveness of BNTX against its peers

3. Specification of industry-specific ratios

1	**Pipeline Maturity Index**	**Why it matters:** Indicates the developmental stage of the pipeline. A higher ratio suggests late-stage readiness and shorter time to market; a lower ratio points to early-stage focus, longer timelines, and higher risk. Useful for benchmarking BioNTech on the R&D maturity curve.
2	**Innovation Momentum**	**Why it matters:** Captures the pace of pipeline expansion and renewal. A high innovation rate suggests strong internal R&D activity, continuous exploration, and a future-looking strategy. It also reflects organizational agility, the ability to rapidly pivot or add new research directions.
3	**R&D Capital Efficiency**	**Why it matters:** Reflects how efficiently BioNTech deploys its R&D budget. Helps benchmark whether the firm is spending in line with its peers relative to pipeline size. A low value (like BioNTech's €43M) can indicate cost-effective innovation, while a high value could suggest capital intensity or inefficiency, depending on pipeline maturity.

BioNTech is **highly competitive as an innovation-led biotech**, excelling in **pipeline expansion** and **cost efficiency**, but still building toward the maturity level of big players like Pfizer and Moderna. **Its competitiveness today lies in its growth potential, agility, and R&D momentum, rather than its current late-stage portfolio**. In 2023, BioNTech's average R&D cost per project was €43 million, not a sign of underinvestment, but a direct outcome of its early-stage pipeline structure. With only 3 out of 41 clinical projects (~7%) in Phase 3, **the company avoids the heavy costs typically concentrated in late-stage development**. This cost structure provides flexibility and supports a high pace of innovation, as reflected by a net pipeline expansion of 9 projects and new trials accounting for 22% of the total pipeline. However, **its long-term competitiveness will depend on whether it can successfully scale this expanding pipeline into late-stage development and marketable therapies**. In other words, BNTX has positioned itself as a future-ready innovator, now it must prove it can translate innovation into late-stage and commercial success.

Source(s): 2023 BioNTech Annual Report, Desk Research

Ratio Analysis – Efficiency and R&D ratios

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Efficiency Ratios		2020	2021	2022	2023	2024
Operating working capital turnover	GSK	-55.00	-25.23	-15.18	-14.12	-11.32
	Peers average	4.41	2.97	1.64	2.99	1.80
Trade receivables turnover	GSK	3.50	3.33	3.93	4.20	4.20
	Peers average	4.79	5.19	5.00	4.68	5.04
Inventories turnover	GSK	1.32	1.39	1.75	1.61	1.47
	Peers average	1.67	1.99	1.86	1.66	1.80
Trade payable turnover	GSK	1.82	1.84	2.27	2.26	2.39
	Peers average	2.83	3.81	3.58	2.77	2.75
Cash conversion cycle	GSK	177.18	171.62	139.17	150.02	179.21
	Peers average	171.97	157.06	160.89	160.31	138.86
Net non-current asset turnover	GSK	0.52	0.52	0.74	0.95	0.86
	Peers average	0.81	0.81	0.78	0.76	0.81

- GSK has a huge different operating working capital turnover compared with its peers due to lower trade receivables, inventories turnover and trade payable turnover than its peers.
- GSK's cash conversion cycle generally longer than its peers, suggesting a slower cash flow cycle and less efficient working capital management.
- GSK has lower net non-current asset turnover than its peers in the past three years due to substantial R&D investment that have not yet to translate to proportional revenue growth rate.

R&D Ratios		2020	2021	2022	2023	2024
R&D Exp/ Rev	GSK	19.68%	20.32%	18.72%	20.52%	20.40%
	Peers average	19.62%	21.01%	20.39%	21.00%	22.11%
RORC (Current Year's Gross Profit/ Previous Year's R&D Exp)	GSK	**	6.86	7.75	7.09	6.50
	Peers average	**	5.67	5.20	5.03	5.36

- Return on R&D Capital (RORC) is consistently higher than peers, suggesting better efficiency in converting R&D spending into gross profit - stronger R&D productivity relative to competitors

**Due to the demerger in 2022, only 2021 and 2020 figures were restated. For comparison, we excluded the 2020 RORC as it requires 2019 data and calculated from the year the figures were restated

Key Forecasts – Revenue

- **2 methods were used to forecast revenue growth (Product-group driven approach and R&D driven approach) and an average of both was taken**

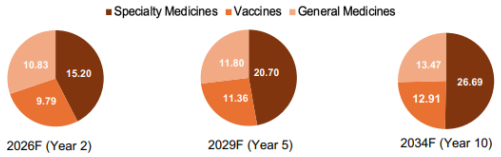
Method 1 : Product-group driven approach - Revenue Forecast Based on Key Drivers

1. **Revenue Breakdown by Product Group**
   Analysed GSK's 3 revenue-generating divisions (Vaccines, Specialty Medicines, and General Medicines) to understand contribution and growth dynamics
2. **Identified Revenue Drivers**
   Selected key products within each segment as primary growth drivers as per GSK's reports (e.g. for Vaccines: Shingles, RSV, Influenza)
3. **Individual Revenue Projections**
   Forecasted future revenue for each key product based on historical performance, guidance, and market trends
4. **Segment Aggregation**
   Summed the revenue across all key products to get total revenue per segment
5. **YOY Growth Rate (%)**
   Calculated year-over-year growth by comparing projected revenue with previous year's actuals

Key Growth Drivers across Portfolios → Sales by Product Groups → Total Sales

- This approach forecasts revenue by identifying key products within GSK's three core segments. By projecting individual product sales and aggregating by segment, it provides a bottom-up view of revenue drivers and allows for the calculation of YoY growth based on underlying performance

Forecasted Revenue by Product Group (£bn)



	Short-term		Medium-term			Long-term				
	2025F	2026F	2027F	2028F	2029F	2030F	2031F	2032F	2033F	2034F
Total Revenue	33.18	35.82	38.21	40.87	43.86	46.19	48.61	50.06	51.56	53.06
YoY growth rate	*11%*	*8%*	*7%*	*7%*	*7%*	*5%*	*5%*	*3%*	*3%*	*3%*

GSK

Based on BNTX's Phase 3 product pipeline as well as comparable commercial products from its competitors, we forecast significant sales growth after 2030

4. Revenue Forecast Evidence

	BNTX Phase 3 Product	The Probability of a High Revenue Source	Comparable Commercialized Drug and Market Revenue
1	Targeted Cancer Antibodies (BNT323/DB-1303 (HER2 ADC)) [1]	• **High potential:** If phase 3 results are strong, then the product could enter a multi-billion-dollar market within 2-3 years • **Fast track:** It received the Fast Track designation and Breakthrough Therapy designations from the FDA for treating endometrial cancer	• AstraZeneca's Enhertu made **$2.6B in 2023 in revenue** and is projected to exceed $10B annually [2]
2	Neoantigen-based T cells: BNT322/GEN1056	• **Moderate potential**: A promising opportunity that needs to demonstrate better safety, stronger efficiency, and differentiation from existing therapies in clinical trials.	• Bristol-Myers Squibb's Yervoy, CTLA-4, generated **$2.24B in revenue in 2023** [3]
3	Covid 19 – Influenza Combination BNT162b2 + BNT161	• **Moderate potential:** Assume that the demand for single shots of COVID-19 and Influenza will shift towards a combination shot.	• The global influenza market size is expected to be at around USD 7.77B in 2023. By 2034, the value is expected to surpass USD 18.02B [4]

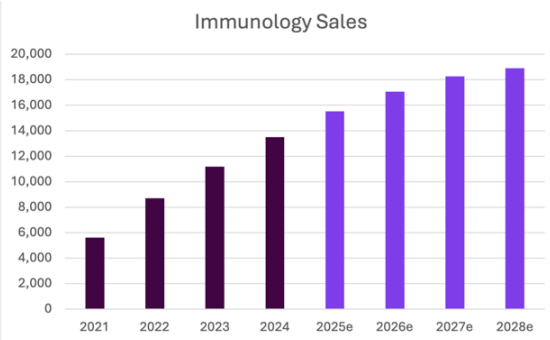
Source(s): (BioNTech, 2024) [1], (Manalac, 2024) [2], (Reuters, 2024) [3] and (BioSpace, 2024) [4]

Prospective Analysis and Key Forecasts - Value drivers

Sanofi value driver: Immunology (Inflammatory) **(EUR€ in millions)**

Expecting **continuous rapid but diminishing** growth rate for Dupixent based on:

1. Widened and stable applicable population due to COPD approval and other therapies.
2. First mover advantage in market share as biologic medication.
3. Underdevelopment of immunology(Inflammation) branch market compared to general immunology(Autoimmune) market.



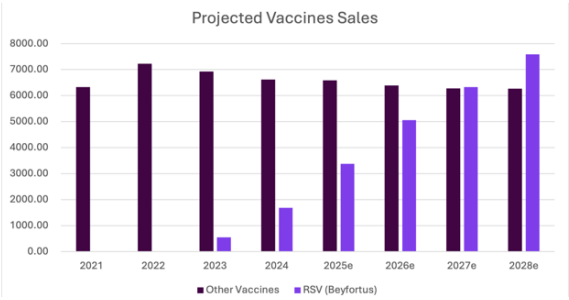
Key forecasting factors

- Projected entire immunology market CAGR around 3-4%
- Immunology section dominated by Dupixent (98% of section sales).
- The long-term/terminal growth rate converge to GDP growth 2.9%.
- Aware of potential future competition of biologic medication of GSK(Nucala) and AstraZeneca(tozorakimab)

Prospective Analysis and Key Forecasts - Value drivers

Sanofi value driver: Vaccines(RSV) **(EUR€ in millions)**

Vaccines	2021	2022	2023	2024	2025e	2026e	2027e	2028e
Vaccines	6323	7229	7474	8298	9957.6	11451.24	12596.364	13856.0004
G rate		14.33%	3.39%	11.02%	20.00%	15.00%	10.00%	10.00%
Breakdown								
Other Vaccines	6,323	7,229	6,927	6,612	6,586	6,393	6,274	6,269
		14.33%	-4.18%	-4.55%	-0.40%	-2.92%	-1.87%	-0.08%
RSV (Beyfortus)			547	1,686	3372	5058	6322.5	7587
				208.23%	100%	50%	25%	20%



Key forecasting factors

1. Expecting Further **penetration** and geographic **expansion**.
2. Stable and abundant market demand.
3. Endow significant but diminishing growth to this specific product in the vaccine section.

Forecasting
Assumptions – Revenue Growth Rate

NOVARTIS

Key Factors Affecting Novartis's Revenue Growth Rate

Global Economic Growth

- Moderate GDP growth and improving healthcare access in emerging markets leads to higher pharmaceutical demand

New Product Launches & Pipeline Maturity

- Biggest upside driver is the launches of new therapies. Successful development leads to immediate revenue acceleration

Lifecycle Management of Existing Products

- New indications, label expansions, or geographic launches of existing drugs can extend revenue growth

M&A and External Innovation

- Recent acquisitions expand Novartis's pipeline and future launch potential. Smart M&A leads to faster access to growth.

Health Care Policy

- pricing is under pressure from policy such as: US Inflation Reduction Act (IRA), European HTA reforms and China volume-based procurement (VBP) etc.

Loss of Exclusivity (LOE)

- Examples: **Entresto** U.S. patent expires 2025
- Generic entry → rapid revenue decline for affected products

Regulatory Approvals & Delays

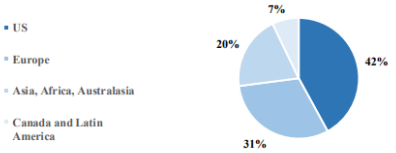
- Faster approvals → growth acceleration
- Delays or rejections by FDA / EMA → risk to forecasted revenue

Currency Exchange Rates

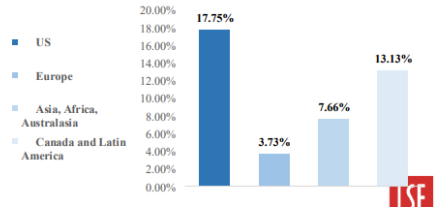
- Novartis reports in USD but earns globally. A strong dollar can negatively impact reported revenue.

Format: Revenue Growth Rate Forecast	202x
Cardiovascular, renal and metabolic	**X**
US (42%)	Ratio 1
Europe (31%)	Ratio 2
Asia, Africa, Australasia (20%)	Ratio 3
Canada and Latin America (7%)	Ratio 4

FY2024 Revenue from continuing operations by region (USD m)



FY2024 Net sales growth rate by region



Forecasting
Assumptions – Revenue Growth Rate

NOVARTIS

- **Forecast continuing operation revenue growth rate**

(growth rate in the rest of the years convert to world GDP growth 2.8%)

Revenue Growth Rate Forecast	2025e	2026e	2027e
Cardiovascular, renal and metabolic	8.08%	6.52%	5.21%
US (42%)	10.00%	7.00%	5.00%
Europe (31%)	6.00%	5.00%	3.00%
Asia, Africa, Australasia (20%)	8.00%	8.00%	9.00%
Canada and Latin America (7%)	6.00%	6.00%	5.00%
Immunology	8.02%	7.75%	7.19%
US (42%)	8.00%	7.00%	6.00%
Europe (31%)	7.00%	7.00%	6.00%
Asia, Africa, Australasia (20%)	10.00%	11.00%	12.00%
Canada and Latin America (7%)	7.00%	6.00%	5.00%
Neuroscience	12.55%	10.86%	9.29%
US (42%)	13.00%	12.50%	10.00%
Europe (31%)	9.00%	7.00%	6.00%
Asia, Africa, Australasia (20%)	18.00%	14.00%	13.00%
Canada and Latin America (7%)	10.00%	8.00%	7.00%
Oncology	13.47%	12.61%	11.66%
US (42%)	15.00%	15.00%	13.00%
Europe (31%)	9.00%	7.00%	7.00%
Asia, Africa, Australasia (20%)	17.00%	16.00%	15.00%
Canada and Latin America (7%)	14.00%	12.00%	12.00%
Established brands	-10.00%	-9.00%	-8.00%
(Expected) Continuing Operations Revenue Growth Rate	5.41%	5.70%	5.69%

- **Forecast other revenue growth rate**

Based on the operation nature and market trend, we assume that other revenue growth will keep stable at 2% for the rest of the years.

- **Derive total revenue growth rate from 2025 to 2034**

	2025e	2026e	2027e	2028e	2029e
(Expected) Continuing Operations Revenue Growth Rate	5.41%	5.70%	5.69%	3.00%	2.70%
(Expected) Total net sales from continuing operations	53,041	56,063	59,253	61,030	62,678
(Expected) Other Revenue Growth Rate	2.00%	2.00%	2.00%	2.00%	2.00%
Other revenue	1,433	1,462	1,491	1,521	1,551
(Expected) Total Revenue	54,474	57,525	60,744	62,551	64,229
(Expected) Total Revenue Growth Rate	5.32%	5.60%	5.60%	2.98%	2.68%

	2030e	2031e	2032e	2033e	2034e
(Expected) Continuing Operations Revenue Growth Rate	2.70%	2.70%	2.70%	2.70%	2.70%
(Expected) Total net sales from continuing operations	64,371	66,109	67,893	69,727	71,609
(Expected) Other Revenue Growth Rate	2.00%	2.00%	2.00%	2.00%	2.00%
Other revenue	1,582	1,614	1,646	1,679	1,713
(Expected) Total Revenue	65,953	67,722	69,540	71,406	73,322
(Expected) Total Revenue Growth Rate	2.68%	2.68%	2.68%	2.68%	2.68%

Novartis management has guided for **mid- to high-single digit** revenue growth, which aligns well with our projection.

ESG ratio overview & BioNTech performance

- BioNTech demonstrates **strong ESG performance and reporting transparency compared to industry peers**, based on analysis of seven self-calculated ESG metrics
- This is in **line with ESG scores provided by 3rd parties** (Morningstar, S&P Global, ISS)

No ESG report published by CureVac

Metric	BIONTECH 2023	BIONTECH 2022	moderna 2023	moderna 2022	Pfizer 2023	Pfizer 2022
CO2e Scope 1+2 emissions [in tons per m$ of revenue]	1.23	0.23	1.75	0.69	18.30	11.17
CO2e Scope 1+2+3 emissions [in tons per m$ of revenue]	140.00	65.17	n.a.	67.91	91.15	104.27
Energy Intensity [MWh/mio USD Rev]	13.6	2.5	15.7	5.4	69.8	42.5
Energy consumption as renewables [in %]	62%	41%	n.a.	n.a.	10%	7%
Total water consumption [per m$ of revenue]	2.88	n.a.	n.a.	n.a.	48.70	39.54
Employee turnover [in %]	6.2%	8.8%	10.0%	9.9%	5.8%	7.3%
Leadership positions occupied by women [in%]	36%	38%	39%	41%	45%	43%

Impact on Analysis

Accounting Analysis
- Requirement to **pay damages from patent infringement case** by Moderna would impact finances[1]
- **Carbon accounting** regulations[2] may ask to account for climate-related risks from high Scope 3 emissions – ESG-linked taxes/pricing schemes may increase costs

Financial & Ratio Analysis
- **Energy efficiency** improvements reduce costs – renewable energy adoption stabilizes future costs but requires upfront capital expenditures
- **Lower borrowing costs** from strong ESG rating may incentivize to take on debt

Strategy Analysis
- Governments & health organizations **may prioritize ESG-conscious firms** in procurement contracts, providing a competitive advantage
- Increased carbon emissions reporting obligations may result in **changes in supply chain strategies**

Prospective Analysis & Valuation
- \+ Strong ESG perf. leads to **higher valuation multiples** due to investor confidence & lower cost of capital
- \- Strong ESG perf. may **reduce profit margins** due to higher costs in the short term - **legal settlements** may result in significant payouts, reducing FCF projections

Note: 1) See appendix for details 2) Such as EUs CSRD
Source(s): 2022 & 2023 Sustainability/ESG reports of BioNTech, Moderna & Pfizer; Morningstar Sustainalytics; S&P Global ESG Score; ISS ESG Score

Environmental, Social, and Governance (ESG) Factors

NOVARTIS

Comparative ESG Analysis in the Pharmaceutical Sector

As of FY 2023	Novartis	Rank	GSK	Rank	Bristol-Myers Squibb	Rank	Johnson & Johnson	Rank	AstraZeneca Avg.	Rank	Eli Lilly and Company	Rank
CO2 Scope 1+2 equivalent emissions (tCO2e) per $ of revenue	6.39	3.00	16.70	6.00	8.15	4.00	5.20	2.00	4.38	1.00	15.44	5.00
CO2 Scope 1+2+3 equivalent emissions (tCO2e) per $ of revenue	107.29	3.00	249.64	6.00	47.06	1.00	85.82	2.00	151.44	5.00	150.61	4.00
Total water consumption (m^3) per $ of revenue	111.44	4.00	191.63	5.00	62.21	2.00	37.58	1.00	78.37	3.00	215.39	6.00
Total energy use (GJ) per $ of Revenue	132.88	4.00	245.74	6.00	118.32	2.00	104.75	1.00	118.77	3.00	223.83	5.00
Recycling intensity (t)	47.27%	4.00	52.72%	1.00	51.62%	2.00	43.18%	5.00	42.00%	6.00	48.00%	3.00
Voluntary Employee Turnover (%)	7.00%	2.00	10.00%	5.00	N/A	6.00	7.40%	3.00	7.50%	4.00	6.07%	1.00
Overall Ranking		3.33		4.83		2.83		2.33		3.67		4.00

NOVARTIS GSK Bristol-Myers Squibb Johnson&Johnson AstraZeneca Lilly

Peers

	Novartis	Pfizer	Roche
MSCI ESG Rating	AA (Leader)	A (Average)	AA
Sustainalytics Risk Score	15.8 (Low Risk)	~19.2 (Low Risk)	~24.3 (Medium Risk)
Access to Medicine Index Rank	#4 (2022)	#6 (2022)	#10 (2022)
DJSI/S&P Global ESG Score	Score 71 (Yearbook member)	Not in top group (flagged for special issues)	Score 79 (Top 2 in sector)
net-Zero Carbon Target	2040 (Scopes 1-3)	2040 (Scopes 1-3)	2050 (Scopes 1-3)
Notable ESG Inclusion	MSCI World ESG Leaders, Sustainalytics Top-Rated	MSCI USA Index; Just 100 (2023)	DJSI World (Top 3 in pharma); FTSE4Good

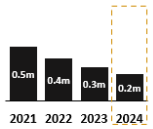
Valuation & Effects

- **Strong ESG:** lower cost of capital (~40 bps difference vs. low ESG peers)
- Investments in sustainability **can slightly affect margins** but mitigate litigation risks
- ESG leadership often priced at a **valuation premium**(lower discount rate, stable growth

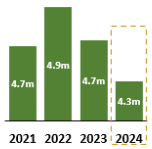
ESG Rankings

- AAA High Performance
- AA Medium High Performance
- B Average Performance
- BB Medium Low Performance
- C Low Performance
- CC Lagging Performance

CO2 Scope 1+2 (tCO2e)



CO2 Scope 3 (tCO2e)



Sources: 1. Annual & ESG Reports of Novartis and respective competitors 2. MSCI ESG Ratings & Sustainalytics 3. S&P Global ESG Scores
Notes:
1. Scope 3 emissions figures may include estimates due to limited disclosure across peer group.
2. "Rank" is calculated on a relative scale (1 = best performer, 6 = laggard) across peers in each ESG metric.